Boot & Execution Flow

Practical Orientation

OS Interface & System Management

BCA 1st Semester

Learning Objectives

- Understand the boot process and system startup sequence
- Learn the distinction between programs and processes
- Explore the concept of services and background operations
- Master OS interface and productivity tools
- Learn to use System Settings for configuration
- Understand Task Manager/Activity Monitor for system monitoring

Boot Process: High Level Overview

Boot Sequence:



Detailed Steps:

- **Power ON:** Electricity reaches the computer
- **BIOS/UEFI:** Firmware initializes hardware (CPU, RAM, Drives)
- **Boot Loader:** Finds and loads the Operating System kernel
- **Kernel Load:** OS core loads into RAM, initializes drivers
- **System Init:** Starts essential services and background processes
- **Ready:** OS displays login screen, system ready for user

Program vs Process

PROGRAM

- **Definition:** Passive set of instructions stored on disk
- **Storage:** Exists as file on hard drive/SSD
- **Execution:** Static until run
- **Example:** application.exe file

PROCESS

- **Definition:** Active instance of program in execution
- **Memory:** Loaded into RAM with allocated resources
- **Execution:** Running, consuming CPU and memory
- **Example:** application.exe running (Process ID: 1234)

Analogy: *A Program is like a Recipe (stored in a book);*
A Process is like someone actively cooking following that recipe.

Introduction to Services

What are Services?

Services are specialized background processes that run automatically with the operating system, often without user interaction, to provide essential system functionality.

Key Characteristics:

- Run in the background automatically
- Provide system-level functionality
- Start when OS boots up
- Do not require user login
- Can be managed through Services console

Common Services:

- Windows Update Service - Updates the OS
- Print Spooler - Manages printing
- Network Service - Enables networking
- Audio Service - Manages sound

OS Interface & Productivity

Primary Components:

Desktop

Visual workspace for files and shortcuts

Taskbar

Quick access to open apps and system tray

File Manager

Browse and manage files and folders

Productivity Tips:

- Use keyboard shortcuts (Win+E for File Manager, Win+D for Desktop)
- Pin frequently used apps to taskbar
- Create desktop shortcuts for quick access
- Organize files in folders by category
- Use virtual desktops for multitasking

System Settings Basics

Accessing System Settings:

Windows: Settings app (Win+I) | macOS: System Preferences | Linux: System Settings

Display
- Resolution
- Brightness
- Refresh Rate

Sound
- Volume
- Input/Output
- Effects

Network
- WiFi
- Ethernet
- VPN

Power
- Sleep Mode
- Hibernation
- Battery

Updates
- OS Updates
- Driver Updates
- Auto-restart

Security
- Firewall
- Antivirus
- Permissions

Task Manager / Activity Monitor Basics

Purpose: *Monitor and manage running processes, system resources, and application performance*

Accessing:

Windows: Ctrl+Shift+Esc or Ctrl+Alt+Del | **macOS:** Cmd+Space, type 'Activity Monitor'

Main Tabs/Views:

Processes/Applications	List of all running programs and their details
Performance	CPU, Memory (RAM), Disk, and Network usage graphs
Startup	Apps that launch automatically at startup
Services	Background services and their status

Common Actions:

- **End Task:** Force close unresponsive applications
- **View Resource Usage:** Monitor CPU, RAM, Disk usage
- **Manage Startup Programs:** Enable/disable auto-startup apps

Key Takeaways

- **Boot Process:** Computer startup involves BIOS → Bootloader → Kernel → System Init → Ready
- **Program vs Process:** Programs are stored files; Processes are active instances in memory
- **Services:** Background processes that provide system functionality automatically
- **OS Interface:** Desktop, Taskbar, and File Manager are primary user interfaces
- **System Settings:** Configure hardware and software parameters for optimal performance
- **Task Manager/Activity Monitor:** Essential tools for monitoring processes and system resources

Thank You!

Questions & Discussion